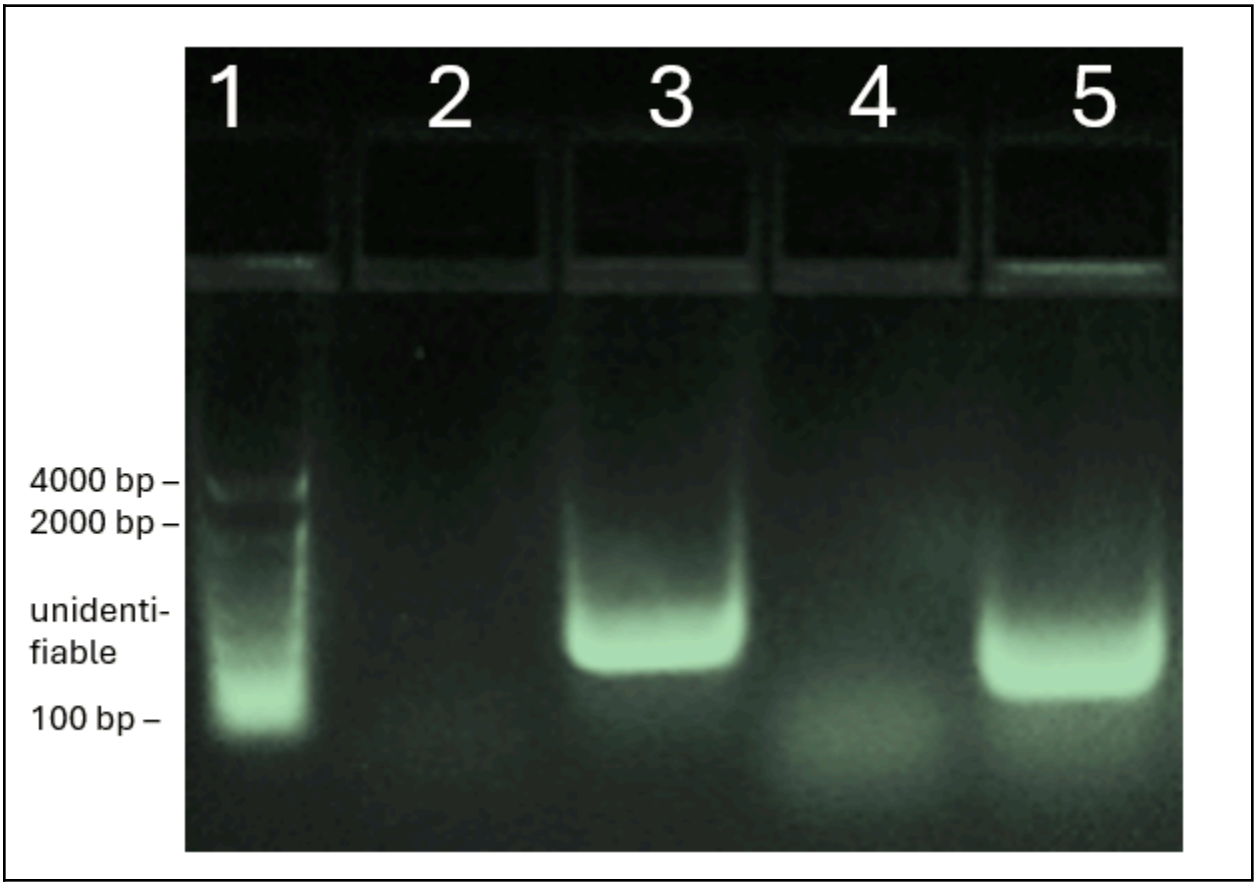


Figure 1. PCR product Gel electrophoresis (Lonza Flash). All lanes went through PCR. Lane 1 is the Lonza DNA Ladder (#50473). Due to smearing, only the 4000bp line could be isolated clearly, and two boundary lines of the smears (2000bp and 100bp) can be identified in the lower half. Additionally, no logarithmic standard curve could be created to align the lengths of the other lanes to the ladder.
Lane 2 is the negative control for the Student Primer, where primers were added to the mastermix but no plasmid was added. Lane 3 is the desired sample for the Student Primer, where primers and plasmid were both added to the mastermix. Lane 4 is the negative control for the Instructor LCT7 Primer, where primers were added to the mastermix but no plasmid was added. Lane 5 is the positive control for the Instructor LCT7 Primer, where primers and plasmid were both added to the mastermix. The plasmid for Lanes 2-5 was the pGEM-T plasmid, and both Student and LCT7 primers were designed to isolate the T7 sequence from the pGEM-T plasmid.

We used flash gels to visualize the successful PCR amplification products. Our intended product was a T7 promoter sequence that would be extracted from the original template plasmid, which previous experiments had verified to be pGEM-T. Primers were designed around the T7 promoter sequence in the pGEM-T plasmid that would anneal to a section of the pGEM-T plasmid and also added in BglII compatible ends for future ligation experiments into our final

cloning vector. PCR was conducted to amplify the proposed T7 insert sequence into a suitable concentration for future ligations.

Our claim is that we successfully amplified the region of choice. We base this claim on the fact that our negative controls- Lane 2 and 4 in Figure 1, both show no clear bands and only smears at the gel front.. We interpret the negative controls as meaning that there is no contamination of external DNA or heterodimer formation of primers within the mastermix used for PCR cycles, as the faint smears represent the initial primers placed into the mixture. We also see that the PCR cycle was calibrated correctly, as our positive control lane 5 in Figure 1, showed that the instructor primers produced a clear band between the DNA ladder boundaries of 2000bp and 100bp. The instructor primers in Lane 5, which have been validated and verified to amplify our pGEM-T sequence correctly, should amplify out a length of 287bp, which is slightly below where our Lane 3, the student primers should be at 430 base pairs. This also supports our claim that our amplified sequence of choice was correct, since DNA at 430bp will migrate slower than DNA at 287bp. However, because the DNA ladder was not clear in the lengths between 100-2000bp, we cannot say with full certainty that our DNA in lanes 3 and 5 were indeed 287bp and 430bp. What we do see is that Lane 3 shows a very clear band that is much smaller than the original plasmid at 3000bp, which means that only one sequence was indeed amplified out correctly from the original plasmid.

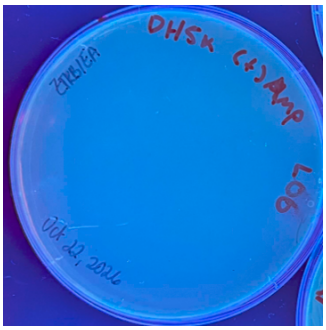
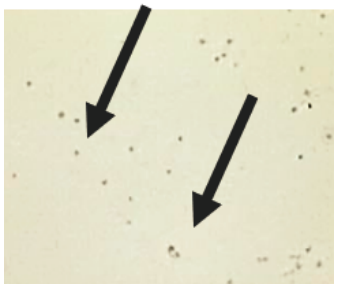
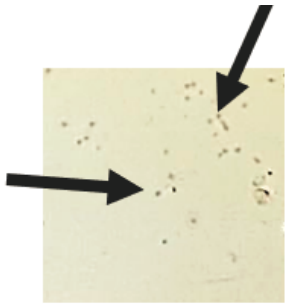
Plate #	1	2	3
Transfected plasmid	No additional plasmid	No additional plasmid	Original GFP-S12 vector
Amp	-	+	+
Plates under UV Light	a 	c 	e 
Close ups of colonies (indicated by arrow)	b 	d  No colonies	f 

Figure 2a-f, remainder of Figure 2 included in the next page

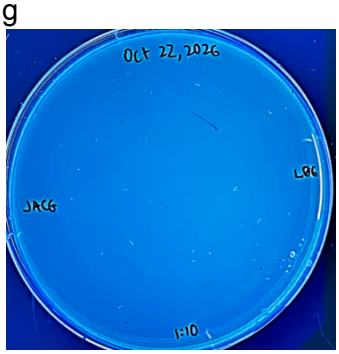
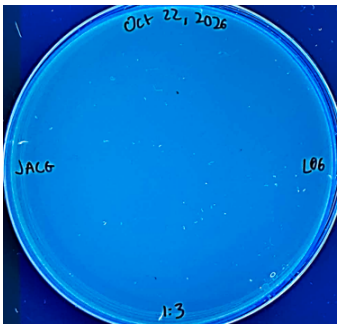
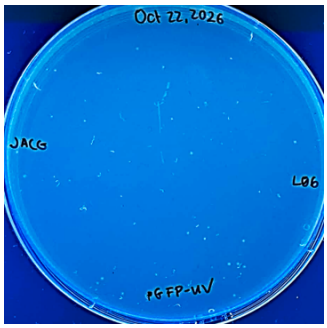
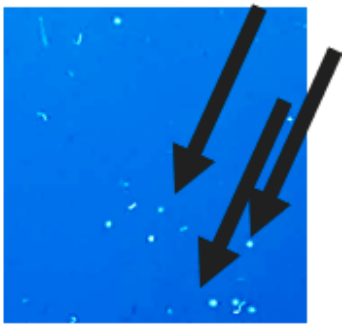
Plate #	4	5	6
Transfected plasmid	Recombinant DNA - T7 insert in pGFP-S12	Recombinant DNA - T7 insert in pGFP-S12	pGFP-UV
Amp	+	+	+
Ligation ratio	1:3 plasmid to insert ratio	1:10 plasmid to insert ratio	N/A
Plates under UV Light	g 	h 	i 
Close ups of colonies (indicated by arrow)	Missing visible light image, but no colonies were formed	Missing visible light image, but no colonies were formed	j 

Figure 2. Transformation of recombinant and control DNA into *E.Coli* alongside controls. All plates were prepped with LB-agar. Additionally, each column of the figure shows a separate treatment on one plate of *E.Coli* competent cells. Plates varied with the type of plasmid (if any) that was transfected into the cells, as well as whether or not Ampicillin (Amp) was mixed into the LB-agar media. Arrows in close-up photos point to the presence of bacterial colonies, but do not point to all existing colonies. Close-ups without arrows indicate absence of bacterial colonies.
Plate 1 was a plate treated with LB-agar and no additional plasmid, and showed growth of bacterial colonies(Figure 2b). None of the colonies fluoresced under UV light (Figure 2a).
Plate 2 was a plate treated with LB-agar-ampicillin and no additional plasmid, and showed no growth of bacterial colonies (Figure 2d). Given that there were no colonies, UV light did not reveal any other growth (Figure 2c).
Plate 3 was a plate treated with LB-agar-ampicillin and the original pGFP-S12 plasmid, and showed growth of bacterial colonies (Figure 2f). None of the colonies fluoresced under UV light (Figure 2e).

Plate 4 was a plate treated with LB-agar-ampicillin and the recombinant pGFP-S12 plasmid ligated at a ratio of 1:3 vector-to-insert, and showed no growth of bacterial colonies (Figure 2g).
Plate 5 was a plate treated with LB-agar-ampicillin and the recombinant pGFP-S12 plasmid ligated at a ratio of 1:10 vector-to-insert, and showed no growth of bacterial colonies (Figure 2h).
Plate 6 was a plate treated with LB-agar-ampicillin and the recombinant pGFP-UV plasmid, and showed both growth of bacterial colonies (Figure 2j). All colonies also displayed fluorescence under UV light (Figure 2i).

After PCR experiments, ligation experiments were carried out to combine the amplified T7 promoter inserts and the host pGFP-S12 vector. To create suitable ligation conditions, all sequences were first digested with BglII to ensure that sticky ends were formed. Then, the insert and vector were incubated at different ratios in combination with T4 Ligase to create a recombinant vector. Finally, transformations into bacteria was conducted.

We claim that the transformations of the recombinant DNA - T7 insert in pGFP-S12 was not successful due to laboratory techniques. This conclusion was determined by examining the expected results of the controls. Plates 1 and 2 represented the control plates where the competent cells were screened for existing ampicillin-resistant genes. We can confidently say that the original competent cells did not have ampicillin-resistant genes because the bacteria grew well in the plate without ampicillin (Plate 1, Figure 2b), but did not grow in the plate with ampicillin (Plate 2, Figure 2d). We also confirmed that the original pGFP-S12 plasmid indeed did have ampicillin in the vector, as we see that the colonies did grow in the presence of ampicillin (Plate 3, Figure 2f). However, in plates 4 and 5, no colonies at all were observed to have grown, with both plates showing a transformation efficiency of 0. Therefore, we conclude that transformation experiments were not successful, as there was no issue with the cells themselves in the control.